

- 1 (a) Fig. 1.1 shows the variation of the length L of an extensible cord with the tension T in the cord.

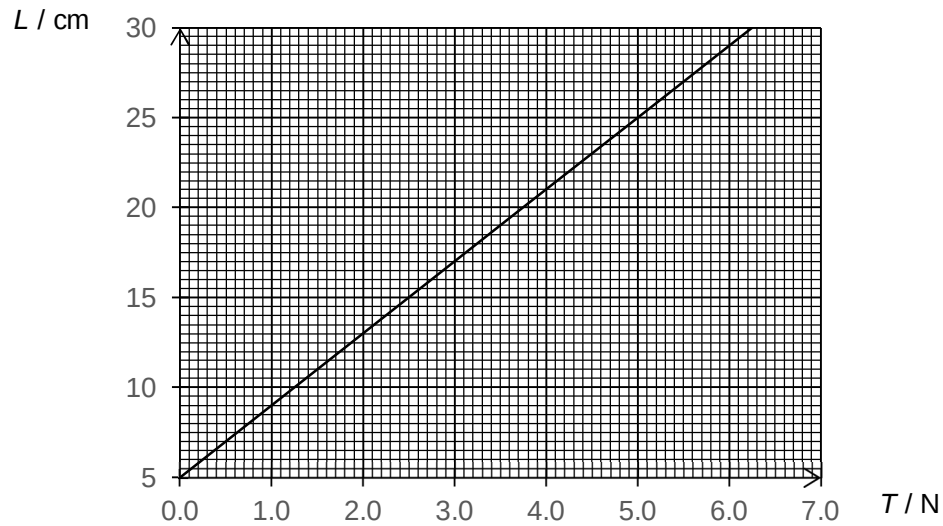


Fig. 1.1

Use Fig. 1.1 to determine

- (i) the spring constant k of the cord,

$$k = \dots\dots\dots \text{N m}^{-1} \quad [2]$$

- (ii) the elastic potential energy in the cord, when the extension of the cord is 20 cm.

$$\text{elastic potential energy} = \dots\dots\dots \text{J} \quad [1]$$

- (b) An L-shaped beam of weight 7.5 N is attached to a wall using a hinge, as shown in Fig. 1.2. The beam is held at rest using the cord in (a), such that the longer part of the beam is horizontal. The weight of the beam acts at a point on the beam, at a horizontal distance x away from the hinge. The extension in the cord is 20 cm.

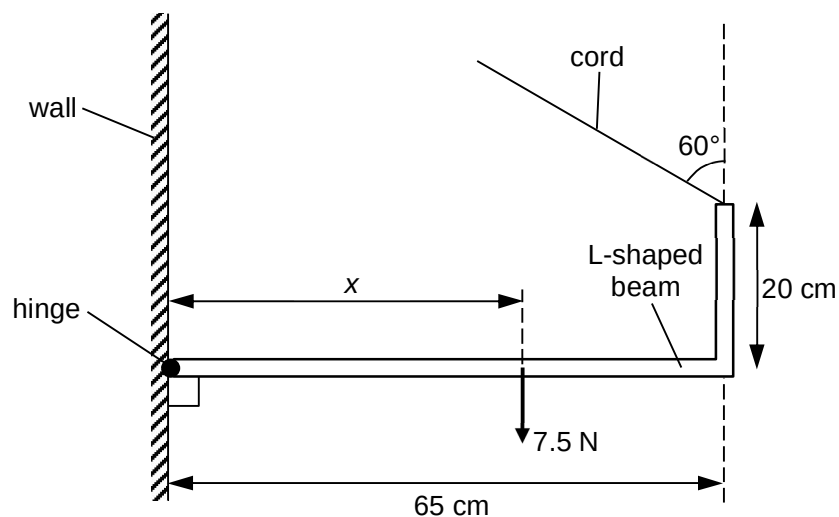


Fig. 1.2

- (i) On Fig. 1.2, draw and label the tension T acted by the cord on the beam, and the reaction force R by the hinge on the beam. [1]
- (ii) Determine the value of x .

$$x = \dots\dots\dots \text{ cm} \quad [3]$$

- (iii) Determine the magnitude of R .

$$R = \dots\dots\dots \text{ N} \quad [2]$$

2 (a) An object P moves at a constant speed v through an arc of a circle of radius r